Advanced Econometrics I
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Problem Set 6—Suggested Answers

Exercises

1. *Setup and key observations.* We are given the potential outcomes framework in regression form:

$$
\begin{aligned}
Y_i &= \beta_0 + \beta_{1i} X_i + u_i,\\
X_i &= \pi_0 + \pi_{1i} Z_i + v_i,
\end{aligned}
$$

with the definitions

$$
\begin{aligned}
\beta_0 &:= \mathrm{E}(Y_i(0)), & \beta_{1i} &:= Y_i(1) - Y_i(0), & u_i &:= Y_i(0) - \mathrm{E}(Y_i(0)),\\
\pi_0 &:= \mathrm{E}(X_i(0)), & \pi_{1i} &:= X_i(1) - X_i(0), & v_i &:= X_i(0) - \mathrm{E}(X_i(0)).
\end{aligned}
$$

The random assignment assumption

$$\big(Y_i(1),\, Y_i(0),\, X_i(1),\, X_i(0)\big) \perp\!\!\!\perp Z_i$$

is the engine of every step below. Because β_{1i}, π_{1i}, u_i, and v_i are all deterministic functions of the potential outcomes $(Y_i(0), Y_i(1), X_i(0), X_i(1))$, they are each individually (and jointly) independent of Z_i.

We will use repeatedly the following consequence of independence:

Lemma 1. *If $A \perp\!\!\!\perp B$ then $\mathrm{E}(AB) = \mathrm{E}(A)\,\mathrm{E}(B)$ for any measurable functions of A and B.*

We also use the shortcut covariance formula $\sigma_{ZW} = \mathrm{E}[(Z_i - \mu_Z) W_i]$ for any random variable W_i (since $\mathrm{E}[(Z_i - \mu_Z)] = 0$).

(a) **Prove that $\sigma_{ZX} = \sigma_Z^2\, \mathrm{E}(\pi_{1i})$.**
Substitute the first-stage equation $X_i = \pi_0 + \pi_{1i} Z_i + v_i$:

$$
\begin{aligned}
\sigma_{ZX} &= \mathrm{E}[(Z_i - \mu_Z)\, X_i]\\
&= \mathrm{E}[(Z_i - \mu_Z)\,(\pi_0 + \pi_{1i} Z_i + v_i)]\\
&= \underbrace{\pi_0\, \mathrm{E}[(Z_i - \mu_Z)]}_{=0} + \mathrm{E}[(Z_i - \mu_Z)\, \pi_{1i} Z_i] + \mathrm{E}[(Z_i - \mu_Z)\, v_i]\,.
\end{aligned}
$$

Third term. Since $v_i \perp\!\!\!\perp Z_i$ and $\mathrm{E}(v_i) = 0$:

$$\mathrm{E}[(Z_i - \mu_Z)\, v_i] = \mathrm{E}(Z_i - \mu_Z)\, \mathrm{E}(v_i) = 0.$$

Second term. Since $\pi_{1i} \perp\!\!\!\perp Z_i$ (by random assignment):

$$\begin{aligned}
\mathrm{E}[(Z_i - \mu_Z)\,\pi_{1i} Z_i] &= \mathrm{E}(\pi_{1i})\,\mathrm{E}[(Z_i - \mu_Z)\,Z_i] \\
&= \mathrm{E}(\pi_{1i})\,\mathrm{E}\big[Z_i^2 - \mu_Z Z_i\big] \\
&= \mathrm{E}(\pi_{1i})\,\big(\mathrm{E}(Z_i^2) - \mu_Z^2\big) = \mathrm{E}(\pi_{1i})\,\sigma_Z^2.
\end{aligned}$$

Collecting terms: $\sigma_{ZX} = \sigma_Z^2\,\mathrm{E}(\pi_{1i})$. □

Intuition. The instrument Z_i shifts X_i only through the individual-specific first-stage slope π_{1i}. The average strength of that shift is $\mathrm{E}(\pi_{1i})$, which is exactly the first-stage coefficient in a linear projection.

(b) **Prove that** $\sigma_{ZY} = \sigma_Z^2\,\mathbf{E}(\beta_{1i}\,\pi_{1i})$**.**
Substitute the structural equation $Y_i = \beta_0 + \beta_{1i} X_i + u_i$:

$$\begin{aligned}
\sigma_{ZY} &= \mathrm{E}[(Z_i - \mu_Z)\,Y_i] \\
&= \underbrace{\beta_0\,\mathrm{E}(Z_i - \mu_Z)}_{=0} + \mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,X_i] + \underbrace{\mathrm{E}[(Z_i - \mu_Z)\,u_i]}_{=0},
\end{aligned}$$

where the third term vanishes by $u_i \perp\!\!\!\perp Z_i$ and $\mathrm{E}(u_i) = 0$, as in part (a).
Now expand X_i in the middle term using the first-stage equation:

$$\begin{aligned}
\mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,X_i] &= \mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,(\pi_0 + \pi_{1i} Z_i + v_i)] \\
&= \pi_0\,\mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}] + \mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,\pi_{1i}\,Z_i] + \mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,v_i]\,.
\end{aligned}$$

First subterm. Since $\beta_{1i} \perp\!\!\!\perp Z_i$:

$$\pi_0\,\mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}] = \pi_0\,\mathrm{E}(Z_i - \mu_Z)\,\mathrm{E}(\beta_{1i}) = 0.$$

Third subterm. The product $\beta_{1i}\,v_i$ is a function of potential outcomes only, so $\beta_{1i}\,v_i \perp\!\!\!\perp Z_i$:

$$\mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,v_i] = \mathrm{E}(Z_i - \mu_Z)\,\mathrm{E}(\beta_{1i}\,v_i) = 0.$$

Second subterm. The product $\beta_{1i}\,\pi_{1i}$ depends only on potential outcomes, hence $\beta_{1i}\,\pi_{1i} \perp\!\!\!\perp Z_i$:

$$\mathrm{E}[(Z_i - \mu_Z)\,\beta_{1i}\,\pi_{1i}\,Z_i] = \mathrm{E}(\beta_{1i}\,\pi_{1i})\,\mathrm{E}[(Z_i - \mu_Z)\,Z_i] = \mathrm{E}(\beta_{1i}\,\pi_{1i})\,\sigma_Z^2.$$

Collecting terms: $\sigma_{ZY} = \sigma_Z^2\,\mathrm{E}(\beta_{1i}\,\pi_{1i})$. □

Putting it together: the LATE. Combining both results:

$$\hat{\beta}^{\mathrm{IV}} = \frac{s_{ZY}}{s_{ZX}} = \frac{\sigma_{ZY}}{\sigma_{ZX}} + \mathrm{o}_p(1) = \frac{\sigma_Z^2\,\mathrm{E}(\beta_{1i}\,\pi_{1i})}{\sigma_Z^2\,\mathrm{E}(\pi_{1i})} + \mathrm{o}_p(1) = \frac{\mathrm{E}(\beta_{1i}\,\pi_{1i})}{\mathrm{E}(\pi_{1i})} + \mathrm{o}_p(1).$$

The probability limit $\mathrm{E}(\beta_{1i}\,\pi_{1i})/\mathrm{E}(\pi_{1i})$ is the *local average treatment effect* (LATE). It is a *weighted* average of the individual treatment effects β_{1i}, with weights proportional to π_{1i}—the strength of the first-stage response for each individual. Compliers (those for whom $\pi_{1i} \neq 0$) receive positive weight; never-takers and always-takers (for whom $\pi_{1i} = 0$) receive zero weight. This is why IV identifies the effect only for the complier subpopulation.

2. *Setup.* We have $Y_i = X_i'\beta + e_i$ with $\mathrm{E}(e_i Z_i) = 0$ (valid instruments) but $\mathrm{E}(e_i X_i) \neq 0$ (endogeneity). The class of estimators indexed by the $K \times L$ matrix P is

$$b_P := \left(\sum_{i=1}^N P Z_i X_i'\right)^{-1} \left(\sum_{i=1}^N P Z_i Y_i\right).$$

Substituting $Y_i = X_i'\beta + e_i$:

$$b_P - \beta = \left(\sum_{i=1}^N P Z_i X_i'\right)^{-1} \sum_{i=1}^N P Z_i e_i = \left(P \cdot \frac{1}{N}\sum_{i=1}^N Z_i X_i'\right)^{-1} P \cdot \frac{1}{N}\sum_{i=1}^N Z_i e_i,$$

where we divided numerator and denominator by N. We use the abbreviations $C_{XZ} = \mathrm{E}(X_i Z_i')$, $C_{ZX} = \mathrm{E}(Z_i X_i')$, $C_{ZZ} = \mathrm{E}(Z_i Z_i')$, and $P^* = C_{XZ} C_{ZZ}^{-1}$.

(a) **Prove that $b_{\hat{P}}$ is consistent.**
By the WLLN:

$$\frac{1}{N}\sum_{i=1}^N Z_i X_i' \xrightarrow{p} C_{ZX}, \qquad \frac{1}{N}\sum_{i=1}^N Z_i e_i \xrightarrow{p} \mathrm{E}(Z_i e_i) = 0,$$

where the last limit is zero by instrument validity. Since $\hat{P} = P^* + \mathsf{o}_p(1)$, Slutsky's theorem and the continuous mapping theorem give:

$$\begin{aligned}
b_{\hat{P}} - \beta &= \left(\hat{P} \cdot \frac{1}{N}\sum_{i=1}^N Z_i X_i'\right)^{-1} \hat{P} \cdot \frac{1}{N}\sum_{i=1}^N Z_i e_i \\
&= ((P^* + \mathsf{o}_p(1))(C_{ZX} + \mathsf{o}_p(1)))^{-1} (P^* + \mathsf{o}_p(1))(\mathrm{E}(Z_i e_i) + \mathsf{o}_p(1)) \\
&= (P^* C_{ZX} + \mathsf{o}_p(1))^{-1} (P^* \cdot 0 + \mathsf{o}_p(1)) \\
&= \left((P^* C_{ZX})^{-1} + \mathsf{o}_p(1)\right) \mathsf{o}_p(1) \qquad P^* C_{ZX} = C_{XZ} C_{ZZ}^{-1} C_{ZX} \\
&= \mathsf{O}_p(1) \cdot \mathsf{o}_p(1) = \mathsf{o}_p(1).
\end{aligned}$$

The key steps are: (i) $P^* C_{ZX}$ is invertible by the relevance assumption (rank condition on the instruments), so the inverse matrix is $\mathsf{O}_p(1)$; and (ii) the right-hand factor is $\mathsf{o}_p(1)$ because $\mathrm{E}(Z_i e_i) = 0$. Hence $b_{\hat{P}} \xrightarrow{p} \beta$. □

Remark. Notice that consistency holds for *any* $\hat{P}$ satisfying $\hat{P} = P^* + \mathsf{o}_p(1)$, not just the optimal choice. Consistency is a weak requirement.

(b) **Derive the asymptotic distribution.**
Multiply the decomposition from part (a) by $\sqrt{N}$:

$$\sqrt{N}(b_{\hat{P}} - \beta) = \underbrace{\left(\hat{P} \cdot \frac{1}{N}\sum_{i=1}^N Z_i X_i'\right)^{-1} \hat{P}}_{=: A_N} \cdot \underbrace{\frac{1}{\sqrt{N}}\sum_{i=1}^N Z_i e_i}_{=: S_N}.$$

Step 1: Limit of A_N. Using $\hat{P} = P^* + \mathrm{o}_p(1)$ and the WLLN:

$$\begin{aligned}
A_N &= ((P^* + \mathrm{o}_p(1))(C_{ZX} + \mathrm{o}_p(1)))^{-1} (P^* + \mathrm{o}_p(1)) \\
&= (P^* C_{ZX} + \mathrm{o}_p(1))^{-1} (P^* + \mathrm{o}_p(1)) \\
&= \left((P^* C_{ZX})^{-1} + \mathrm{o}_p(1)\right) (P^* + \mathrm{o}_p(1)) \\
&= (P^* C_{ZX})^{-1} P^* + \mathrm{o}_p(1).
\end{aligned}$$

Step 2: CLT for S_N. By the CLT and instrument validity $\mathrm{E}(Z_i e_i) = 0$:

$$S_N = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} Z_i e_i \xrightarrow{d} \mathrm{N}\big(0,\ \mathrm{E}(e_i^2 Z_i Z_i')\big) = \mathrm{N}\big(0,\ \sigma_e^2\, C_{ZZ}\big)\,,$$

where the simplification uses the homoskedasticity assumption $\mathrm{E}(e_i^2 Z_i Z_i') = \sigma_e^2 C_{ZZ}$.
Step 3: Slutsky. Since $A_N \xrightarrow{p} (P^* C_{ZX})^{-1} P^*$ and $S_N \xrightarrow{d} \mathrm{N}(0, \sigma_e^2 C_{ZZ})$:

$$\sqrt{N}(b_{\hat{P}} - \beta) \xrightarrow{d} \mathrm{N}\Big(0,\ \sigma_e^2\, (P^* C_{ZX})^{-1} P^*\, C_{ZZ}\, P^{*\prime} \big[(P^* C_{ZX})^{-1}\big]'\Big).$$

Step 4: Simplify with $P^* = C_{XZ} C_{ZZ}^{-1}$.
Denote $A := P^* C_{ZX} = C_{XZ} C_{ZZ}^{-1} C_{ZX}$. Note that A is symmetric because C_{ZZ} is symmetric and $C_{XZ} = C_{ZX}'$:

$$A' = C_{ZX}' (C_{ZZ}^{-1})' C_{XZ}' = C_{XZ} C_{ZZ}^{-1} C_{ZX} = A.$$

Compute the sandwich matrix:

$$\begin{aligned}
(P^* C_{ZX})^{-1} P^*\, C_{ZZ}\, P^{*\prime} \big[(P^* C_{ZX})^{-1}\big]' &= A^{-1}\, \underbrace{C_{XZ} C_{ZZ}^{-1}}_{= P^*}\, C_{ZZ}\, \underbrace{C_{ZZ}^{-1} C_{ZX}}_{= P^{*\prime}}\, A^{-1} \\
&= A^{-1}\, C_{XZ}\, \underbrace{C_{ZZ}^{-1} C_{ZZ} C_{ZZ}^{-1}}_{= C_{ZZ}^{-1}}\, C_{ZX}\, A^{-1} \\
&= A^{-1}\, \underbrace{C_{XZ} C_{ZZ}^{-1} C_{ZX}}_{= A}\, A^{-1} \\
&= A^{-1}.
\end{aligned}$$

Therefore, the asymptotic distribution simplifies to:

$$\sqrt{N}(b_{\hat{P}} - \beta) \xrightarrow{d} \mathrm{N}\big(0,\ \sigma_e^2\, (C_{XZ} C_{ZZ}^{-1} C_{ZX})^{-1}\big).$$

Remark. This is the textbook 2SLS asymptotic variance under homoskedasticity. The remarkable simplification—from a complicated sandwich to a clean inverse—occurs precisely because P^* is the efficient weighting matrix. Any other P would yield a larger asymptotic variance (in the positive semi-definite sense), which is why P^* is called the *optimal* weight matrix. The estimator $b_{\hat{P}}$ with this choice is the two-stage least squares (2SLS) estimator.

(c) **Suggest a consistent estimator $\hat{P}$ for P^*.**

analog principle

We need $\hat{P} = C_{XZ}C_{ZZ}^{-1} + o_p(1)$. The WLLN gives

$$\frac{1}{N}\sum_{i=1}^{N} X_i Z_i' \xrightarrow{p} \mathrm{E}(X_i Z_i') = C_{XZ}, \qquad \frac{1}{N}\sum_{i=1}^{N} Z_i Z_i' \xrightarrow{p} \mathrm{E}(Z_i Z_i') = C_{ZZ}.$$

By the continuous mapping theorem (matrix inversion is continuous at invertible matrices):

$$\hat{P} := \left(\frac{1}{N}\sum_{i=1}^{N} X_i Z_i'\right)\left(\frac{1}{N}\sum_{i=1}^{N} Z_i Z_i'\right)^{-1} \xrightarrow{p} C_{XZ}C_{ZZ}^{-1} = P^*.$$

This is simply the OLS estimator from a multivariate regression of X_i on Z_i, which is the first-stage regression of 2SLS.

Summary. Questions 2(a)–(c) walk through the logic of 2SLS step by step: (a) any consistent preliminary estimator of P^* preserves consistency; (b) the optimal P^* yields the smallest asymptotic variance $\sigma_e^2(C_{XZ}C_{ZZ}^{-1}C_{ZX})^{-1}$; (c) the sample analog $\hat{P}$ is the first-stage OLS fit. Together these form the two-stage structure of 2SLS.